§2.2: andwiched Singularities, Extended Graphs, and Decorated Germs

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1. Reduced Fundamental Cycle

Definition 1.1. $(X, 0)$ is a *rational singularity with reduced fundamental cycle* if it admits a normal crossing resolution such that all exceptional curves have genus 0, the dual resolution graph G is a tree, and for each vertex $v \in G$, the valency $a(v)$ of v and the self-intersection $v \cdot v$ satisfy the inequality

$$a(v) \leq -v \cdot v. \tag{1}$$

Recall that the *fundamental cycle* $Z = \sum_{i=1}^{n} c_i E_i$ of a rational surface singularity is the unique minimal positive divisor supported on the exceptional curves satisfying the anti-nef condition:

$$Z \cdot E_i \leq 0 \quad \text{for all } i = 1, \ldots, n. \tag{2}$$

A singularity has a *reduced fundamental cycle* if every coefficient $c_i = 1$, so that $Z = \sum_{j=1}^{n} E_j$.

We can directly verify the numerical condition in Definition 2.1 by expanding the intersection product for a given exceptional curve E_i (represented by a vertex $v \in G$):

$$Z \cdot E_i = \left(\sum_{j=1}^{n} E_j \right) \cdot E_i = E_i \cdot E_i + \sum_{j \neq i} E_j \cdot E_i. \tag{3}$$

In a normal crossings resolution, $E_i \cdot E_i$ is the self-intersection number $v \cdot v$, and the sum $\sum_{j \neq i} E_j \cdot E_i$ counts the number of intersections with other exceptional curves, which is precisely the valency $a(v)$ of the vertex in the dual graph G.

Thus, the anti-nef condition becomes:

$$v \cdot v + a(v) \leq 0 \implies a(v) \leq -v \cdot v, \tag{4}$$

recovering the exact inequality given in the paper.

2. Sandwiched Singularities

de Jong and van Straten state the following simple definition regarding a sandwiched singularity.

Definition 2.1. A *sandwiched singularity* is, by definition, a normal surface singularity that admits a birational map to $(\mathbb{C}^2, 0)$.

To unpack the definition of a sandwiched singularity $(X, 0)$, the geometric construction works by starting with the smooth flat plane $\mathbb{C}^2$ and building the singularity through a three-step process:

$$\tilde{X} \xrightarrow{\text{contract curves}} X \xrightarrow{\text{map to plane}} \mathbb{C}^2 \tag{5}$$

First, we start with $(\mathbb{C}^2, 0)$ and perform a sequence of point blowups to create a smooth resolved surface $\tilde{X}$. This process populates $\tilde{X}$ with a collection of intersecting exceptional curves.

Second, we choose a specific subset of these exceptional curves to pinch or collapse down to a single sharp point.

Third, this contracting process yields our singular surface X. Because X sits directly between the smooth resolution $\tilde{X}$ and the flat plane $\mathbb{C}^2$, it is called a sandwiched singularity. The exceptional curves that were not collapsed project down into $\mathbb{C}^2$ as an arrangement of plane curves, allowing us to study the deformations of the singularity by looking at how these flat curves shift and smooth out.

The property of being a sandwiched singularity is entirely encoded within the dual resolution graph G. While finding the embedding graph G' can feel like a game of trial and error, the reduced fundamental cycle (RFC) condition provides a formal, deterministic method to construct it.

For each vertex $v \in G$, the RFC inequality $a(v) \leq -v \cdot v$ implies that:

$$k_v = -v \cdot v - a(v) \geq 0. \tag{6}$$

The integer k_v represents the exact number of new (-1)-valued vertices (leaves) that must be attached to v via a single edge.

By attaching exactly k_v separate (-1) leaves to every vertex $v \in G$, we systematically construct the expanded graph G'. When these new (-1) leaves are blown down one by one, each blowdown adds $+1$ to the self-intersection of its host vertex v. Once all added leaves are collapsed, the self-intersection of every original vertex increases from $v \cdot v$ to:

$$v \cdot v + k_v = -a(v). \tag{7}$$

This perfectly balances the remaining weights against the tree's valencies, allowing the entire remaining tree to cleanly collapse to a single smooth point (S^3 boundary). This proves that every rational singularity with a reduced fundamental cycle is automatically sandwiched.

3. Curvettas

Every sandwiched singularity is associated with a plane curve arrangement $C \subset \mathbb{C}^2$ via a geometric construction called a *curvetta*.

Once the expanded graph G' is chosen, it represents a smooth surface containing a specific arrangement of rational curves. For each of the distinguished (-1) curves added to form G', we choose a small transverse complex disk-called a **curvetta** intersecting that curve at a generic point.

When we contract the entire curve configuration down to the flat plane $\mathbb{C}^2$, the (-1) curves collapse, but the transverse curvettas survive. They map directly down to the origin to form the irreducible components C_i of a reducible plane curve $C \subset \mathbb{C}^2$. For rational singularities with a reduced fundamental cycle, each component C_i maps down as a completely smooth disk at the origin.

Definition 3.1. Let $\rho : \tilde{Z} \to Z = (\mathbb{C}^2, 0)$ be a sequence of point blow-ups with the exceptional set $F := \rho^{-1}(0)$.

Definition 3.2. The *sandwiched singularity* X, determined by $\rho : \tilde{Z} \to Z$, is obtained by contracting the set E of all non-(-1)-curves of $\tilde{Z}$. (We assume for now that this configuration is connected.)

4. Decorated Germs

Each curvetta C_i comes with a weight $w_i = w(C_i)$, given by the number of exceptional spheres that intersect the corresponding curve in the blow-down process from G' to the empty graph. In other words, w_i is the number of blow-down steps that affect the corresponding curvetta before it becomes C_i. The weighted curve $(\mathcal{C}, w)$ is called a *decorated germ* corresponding to $(X, 0)$.

Definition 4.1. For a plane curve germ $C = \bigcup_{i \in T} C_i$, we define the following numbers:

1. $m(i)$ is the sum of multiplicities of branch i in the multiplicity sequence of the minimal resolution of C.

2. $M(i)$ is the sum of multiplicities of branch i in the multiplicity sequence of the minimal good resolution of C.

Definition 4.2. A *decorated curve* is a pair (C, l) consisting of:

1. a curve singularity $C = \bigcup_{i \in T} C_i \subset (\mathbb{C}^2, 0)$,
2. a function $l : T \to \mathbb{Z}$ assigning to each branch of C a number,
3. with the condition that $l(i) \geq m(i)$.

Definition 4.3. Let (C, l) be a decorated curve.

1. The modification $\tilde{Z}(C, l) \to Z$ determined by (C, l) is obtained from the minimal embedded resolution of C by $l(i) - m(i)$ consecutive blow-ups at the i-th branch of C.
2. The analytic space $X(C, l)$ is obtained from $\tilde{Z}(C, l) - \tilde{C}$ by blowing down the maximal compact set, that is, the union of all exceptional divisors not intersecting the strict transform $\tilde{C} \subset \tilde{Z}(C, l)$.

References

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